

Math 142, S12

Quiz 3

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find

$$\int \frac{1}{x^2 \sqrt{16 - x^2}} dx$$

using the substitution $x = 4 \sin \theta$.

Solution: Let $x = 4 \sin \theta$, then $dx = 4 \cos \theta d\theta$.

$$\int \frac{1}{x^2 \sqrt{16 - x^2}} dx = \int \frac{1}{(4 \sin \theta)^2 \sqrt{16 - (4 \sin \theta)^2}} \cdot 4 \cos \theta d\theta$$

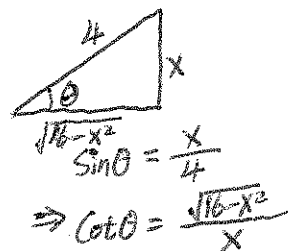
$$= \int \frac{4 \cos \theta}{16 \sin^2 \theta \sqrt{16(1 - \sin^2 \theta)}} d\theta$$

$$= \int \frac{\cancel{4} \cos \theta}{16 \sin^2 \theta \cancel{4} \cos \theta} d\theta$$

$$= \int \frac{1}{16} \csc^2 \theta d\theta$$

$$= -\frac{1}{16} \cot \theta + C$$

$$= -\frac{\sqrt{16 - x^2}}{16x} + C$$



Problem 2. (5 points) Evaluate

$$\int \frac{x-9}{(x+5)(x-2)} dx$$

using the partial fraction decomposition.

Solution: We know

$$\begin{aligned} \frac{x-9}{(x+5)(x-2)} &= \frac{A}{x+5} + \frac{B}{x-2} \\ &= \frac{A(x-2) + B(x+5)}{(x+5)(x-2)} \\ &= \frac{(A+B)x - 2A + 5B}{(x+5)(x-2)} \end{aligned}$$

$$\Rightarrow \begin{cases} A+B=1 \\ -2A+5B=-9 \end{cases} \Rightarrow \begin{matrix} A=2 \\ B=-1 \end{matrix}$$

Thus

$$\begin{aligned} \int \frac{x-9}{(x+5)(x-2)} dx &= \int \left(\frac{2}{x+5} - \frac{1}{x-2} \right) dx \\ &= 2 \ln|x+5| - \ln|x-2| + C \end{aligned}$$